

OPERATING SYSTEM ADMINISTRATION & ACCESS CONTROL STRUCTURE

IT Support Professional
Certificate Capstone Project

PREPARED BY:

The Pink Black Panthers

29/05/2026

Tech Career Accelerator
Google IT Support Capstone Project

Table of Contents

Table of Contents	1
Introduction	2
User and Permissions Matrix.....	3
Shared Folder Design.....	4
The Public Folder	4
The Finance Folder	4
The HR Folder	4
The Sales Folder	5
The Development Folder.....	5
The Management Folder	5
Access Control Structure	6
Command-Line Administration Guide.....	7
Software Deployment Plan	10
Standard Software for All Employees.....	10
Finance Department Software Deployment.....	11
Human Resources Department Software Deployment	11
Sales and Marketing Department Software Deployment	12
Software Development Department Software Deployment	12
IT Support and Administration Tools	13
Software Installation Procedures	14
Software Update and Patch Management.....	14
Licensing Management.....	15
IP Addressing Plan.....	16
Benefits of the Proposed Design	17
Conclusion	18

Introduction

Ubuntu Innovations (Pty) Ltd requires a structured user access and file management system to ensure employees can securely access the resources necessary for their job functions. Proper user permissions and folder access controls help protect sensitive company information while improving collaboration and operational efficiency.

This document outlines the proposed User and Permissions Matrix and Shared Folder Design for the organization.

User and Permissions Matrix

To improve security and enforce access control, employees will be assigned to security groups based on their department. Access to company folders and resources will be granted according to job responsibilities.

Department	Security Group	Folder Access
Management	mgmt._grp	Full Access
Finance	finance_grp	Finance Folder
HR	hr_grp	HR Folder
Sales	sales_grp	Sales Folder
Development	dev_grp	Developmental Folder

Shared Folder Design

A structured shared folder environment will be implemented to organize company information and ensure appropriate access controls.

/Company

|

|—— *Public*

|—— *Finance*

|—— *HR*

|—— *Sales*

|—— *Development*

|—— *Management*

The Public Folder

The Public folder contains information that can be accessed by all employees, including company announcements, templates, and shared resources.

The Finance Folder

The Finance folder contains financial records, budgets, invoices, and accounting documentation. Access is restricted to authorized finance personnel and management.

The HR Folder

The HR folder contains employee records, recruitment documents, training records, and human resource information. Access is restricted to HR staff and management.

The Sales Folder

The Sales folder stores customer information, sales reports, quotations, and sales-related documentation. Access is limited to the sales department and management.

The Development Folder

The Development folder contains project documentation, source code repositories, technical resources, and development-related information. Access is limited to the development team and management.

The Management Folder

The Management folder contains strategic planning documents, executive reports, financial summaries, and confidential organizational information. Access is restricted to management personnel only.

Access Control Structure

The proposed folder structure will use role-based access control (RBAC) principles to manage permissions. Employees will only receive access to folders required for their job responsibilities.

Access rights will be assigned through security groups rather than individual user permissions. This simplifies administration and improves security management.

Folder	Access Granted To
Public	All Employees
Finance	Finance + Management
HR	HR + Management
Sales	Sales + Management
Development	Developmental + Management
Management	Management Only

Command-Line Administration Guide

This guide explains basic Linux administration commands used in Ubuntu systems. These commands are useful for IT support, software administration, system maintenance, and development tasks used at Ubuntu Innovations (Pty) Ltd. The company uses tools such as Python, PostgreSQL, GitHub, Wireshark, and Windows/Ubuntu administration software from the software inventory.

Command: *pwd*

Purpose: Displays the current working directory in the Linux file system.

Example: *pwd*

Why It's Useful: Helps users know exactly where they are working in the system.

Simple Explanation: It shows the folder you are currently inside.

Command: *ls*

Purpose: Lists files and folders in the current directory.

Example: *ls*

Why It's Useful: Allows administrators and developers to view files, scripts, and software folders quickly.

Simple Explanation: It shows what files and folders are in the current location.

Command: *cd*

Purpose: Changes from one directory to another.

Example: *cd Documents*

Why It's Useful: Helps users move around the Linux file system to access software files, logs, or projects.

Simple Explanation: It lets you move into another folder.

Command: *mkdir*

Purpose: Creates a new directory (folder).

Example: *mkdir* Projects

Why It's Useful: Useful for organising software projects, backups, and administrative files.

Simple Explanation: It creates a new folder.

Command: *touch*

Purpose: Creates a new empty file.

Example: *touch* notes.txt

Why It's Useful: Useful for creating configuration files, scripts, or documentation files.

Simple Explanation: It creates a new blank file.

Command: *chmod*

Purpose: Changes file or folder permissions.

Example: *chmod* 755 script.sh

Why It's Useful: Controls who can read, write, or execute files. This improves system security.

Simple Explanation: It changes what users are allowed to do with a file.

Command: *chown*

Purpose: Changes the owner of a file or directory.

Example: *sudo chown* laeeq:laeeq report.txt

Why It's Useful: Helps administrators manage file ownership and user access.

Simple Explanation: It changes who owns the file.

Command: *ps*

Purpose: Displays currently running processes.

Example: *ps*

Why It's Useful: Helps administrators monitor active programs and troubleshoot problems.

Simple Explanation: It shows which programs are currently running.

Command: *top*

Purpose: Shows real-time system processes and resource usage.

Example: *top*

Why It's Useful: Useful for monitoring CPU usage, memory usage, and system performance.

Simple Explanation: It gives a live view of what the computer is doing.

Command: *apt install*

Purpose: Installs software packages on Ubuntu Linux.

Example: *sudo apt install python3*

Why It's Useful: Allows administrators to install software tools such as Python, PostgreSQL, Wireshark, and other applications used by Ubuntu Innovations.

Simple Explanation: It downloads and installs software on Ubuntu.

These Linux commands are important for beginner system administrators, developers, and IT support staff. They help users manage files, monitor systems, install software, and maintain secure and organised computer systems. Many of these commands support the software and administration tasks required at Ubuntu Innovations (Pty) Ltd.

Software Deployment Plan

The software deployment plan outlines the procedures and software solutions required to support the operations of Ubuntu Innovations (Pty) Ltd. The deployment strategy ensures that employees have access to standardized, secure, and updated software applications necessary for productivity, communication, administration, software development, and cybersecurity. The deployment plan also supports software management, licensing compliance, system maintenance, and business continuity across all departments.

Standard Software for All Employees

The following software applications will be deployed to all employee workstations to support daily business operations, communication, collaboration, security, and internet access.

Software	Users	Deployment method	License Type	Update Method	Purpose
Windows 11 Pro	25	Pre-installed on company devices	Per-Device License	Automatic Windows Updates	Operating system management and device security
Microsoft 365	25	Cloud-based deployment through company accounts	Business Subscription	Automatic Cloud Updates	Email, document creation, collaboration, presentations, and cloud storage
Microsoft Defender	25	Enabled during workstation configuration	Included with Windows	Automatic Security Updates	Antivirus protection and endpoint security
Google Chrome / Microsoft Edge	25	Installed during workstation setup	Free	Automatic Browser Updates	Internet access and web-based applications
Adobe Acrobat Reader	25	Installed during workstation setup	Free	Automatic Updates	Viewing and printing PDF documents

Microsoft Entra ID (Azure AD)	25	Configured through company domain accounts	Business Subscription	Managed Cloud Updates	User authentication and access control
-------------------------------	----	--	-----------------------	-----------------------	--

Finance Department Software Deployment

The Finance Department requires software applications that support payroll administration, accounting operations, budgeting, and financial reporting.

Software	Users	Deployment method	License Type	Update Method	Purpose
Sage Accounting & Payroll	4	Cloud-based deployment through secure company accounts	Business Subscription	Automatic Cloud Updates	Financial management, payroll processing, and tax calculations

Human Resources Department Software Deployment

The Human Resources Department requires software solutions that support employee management, leave administration, recruitment, and internal communication.

Software	Users	Deployment method	License Type	Update Method	Purpose
BambooHR	3	Cloud-based deployment through company accounts	Business Subscription	Automatic Cloud Updates	Employee records, leave management, and recruitment tracking

Sales and Marketing Department Software Deployment

The Sales and Marketing Department requires software applications that support customer engagement, campaign management, branding, and digital content creation.

Software	Users	Deployment method	License Type	Update Method	Purpose
Canva Pro	7	Cloud-based deployment through team accounts	Team Subscription	Automatic Cloud Updates	Marketing material and presentation design
HubSpot CRM	7	Cloud-based deployment through company accounts	Subscription	Automatic Cloud Updates	Customer relationship management and sales tracking

Software Development Department Software Deployment

The Software Development Department requires specialized software tools for application development, database management, version control, debugging, and collaboration.

Software	Users	Deployment method	License Type	Update Method	Purpose
Visual Studio Code	8	Installed on development workstations	Free / Open Source	Automatic Updates	Software development and debugging
GitHub Desktop	8	Installed on development workstations	Free	Automatic Updates	Version control and code collaboration
Python	8	Installed on development workstations	Free / Open Source	Manual and Automatic Updates	Application development and scripting
MySQL Workbench	8	Installed on development workstations	Free / Open Source	Automatic Updates	SQL development and database administration

PostgreSQL	8	Installed on development workstations	Free / Open Source	Scheduled Updates	Relational database management
MongoDB Compass	8	Installed on development workstations	Free	Automatic Updates	NoSQL database management
GitHub Copilot	8	Configured through GitHub accounts	Subscription	Managed Cloud Updates	AI-assisted coding and productivity
ChatGPT	8	Cloud-based access through company accounts	Subscription / Free	Managed Cloud Updates	AI-assisted development and documentation support

IT Support and Administration Tools

The following software solutions support system administration, troubleshooting, backup management, and technical support operations within the organisation.

Software	Users	Deployment Method	License Type	Update Method	Purpose
TeamViewer	3	Installed on IT support devices	Subscription	Automatic Updates	Remote technical support and troubleshooting
Jira Service Management	3	Cloud-based deployment through IT accounts	Subscription	Automatic Cloud Updates	Ticket and incident management
Wireshark	3	Installed on IT administration devices	Free / Open Source	Manual Updates	Network troubleshooting and diagnostics
Veeam Backup & Recovery	3	Installed on backup management systems	Subscription	Scheduled Updates	Data backup and disaster recovery

Software Installation Procedures

The following procedures will be followed during software deployment within Ubuntu Innovations (Pty) Ltd:

1. Company devices will be configured with Windows 11 Pro before employee allocation.
2. Standard software applications will be installed on all devices during workstation setup.
3. Department-specific software will be deployed according to employee job roles and responsibilities.
4. Employees will receive company-managed user accounts and login credentials.
5. Cloud-based software platforms will be configured using company email accounts.
6. IT support staff will verify software functionality before devices are issued to employees.
7. Software installations and licenses will be documented for future maintenance and auditing purposes.

Software Update and Patch Management

To ensure system security, reliability, and performance, software updates will be managed through the following procedures:

- Automatic Windows updates will be enabled on all company devices.
- Antivirus definitions and security patches will update automatically.
- Cloud-based software applications such as Microsoft 365 and BambooHR will receive vendor-managed updates.
- Critical security updates will be deployed immediately by IT support staff.
- Unsupported or outdated software will be removed from company systems.
- Scheduled maintenance checks will be performed to verify software compliance and update status.

Licensing Management

Ubuntu Innovations (Pty) Ltd will maintain valid software licenses and subscriptions for all commercial software used within the organisation.

Commercial software licenses include:

- Microsoft 365 Business
- Windows 11 Pro
- Sage Accounting & Payroll
- Canva Pro
- HubSpot CRM
- GitHub Copilot
- TeamViewer
- Jira Service Management
- Veeam Backup & Recovery

Free and open-source software applications such as Visual Studio Code, Python, PostgreSQL, Wireshark, and GitHub Desktop may be used without commercial licensing costs where applicable.

The software deployment plan provides Ubuntu Innovations (Pty) Ltd with a structured approach for deploying, managing, updating, and securing software applications across all departments. The deployment procedures support productivity, cybersecurity, collaboration, software compliance, and operational efficiency while ensuring that employees have access to the tools required for their specific business functions.

IP Addressing Plan

Device Category	Network Address (CIDR)	Usable Host Range	Subnet Mask	Description / Purpose
Servers	192.168.10.0/24	192.168.10.1 - 192.168.10.254	255.255.255.0	Critical infrastructure, active directory, databases, and local applications. Typically assigned static IPs.
User Devices	192.168.20.0/24	192.168.20.1 - 192.168.20.254	255.255.255.0	Wired corporate workstations, desktop PCs, and employee laptops. Dynamically assigned via DHCP.
Printers	192.168.30.0/24	192.168.30.1 - 192.168.30.254	255.255.255.0	Networked multi-function printers and copiers. Assigned static IPs or reserved DHCP addresses.
Wi-Fi	192.168.40.0/24	192.168.40.1 - 192.168.40.254	255.255.255.0	Wireless access points handling mobile devices, guest access, and wireless corporate assets.

The proposed IP addressing scheme separates network resources into dedicated subnets to improve security, simplify network management, and support future scalability.

Servers are isolated from user devices, while printers and wireless devices operate on separate subnets to reduce unnecessary network traffic and improve performance. This structure also simplifies troubleshooting and access control.

Benefits of the Proposed Design

The proposed user permission and folder structure provides several benefits:

- Improved security through restricted access controls.
- Protection of sensitive business information.
- Simplified user management through security groups.
- Reduced risk of accidental data exposure.
- Better organization of company resources.
- Easier administration and scalability as the company grows.

Conclusion

The implementation of a User and Permissions Matrix combined with a structured Shared Folder Design will help Ubuntu Innovations (Pty) Ltd maintain secure access to company information. By utilizing security groups and role-based access controls, the organization can improve data protection while supporting efficient collaboration between departments.
